Modeling and Validation of a Soccer Ball's Three-Dimensional Flight to the Goal

Problem B

Abstract

Gravity bends a football's path downward, drag reduces its speed, and Magnus force redirects a spinning flight. We compare gravity, gravity with drag, a constant resultant force, and gravity with aerodynamic forces. M3 determines constant acceleration from three early positions. M4 uses constant drag and the linear lift relation $C_L = \alpha Sp$ to estimate velocity, spin direction and aerodynamic parameters.

Three actual penalties supply 39 sampled frames. Three early positions per flight enter inversion, with subsequent measurements used for comparison. M4 gives goal-plane errors of 0.360, 0.146 and 0.113 m for the two placement shots and the power shot; the latter has a subsequent path RMSE of 0.093 m. A separate competition-position example quantifies parameter effects: a 1° direction change moves the crossing by about 0.20 m, and the largest grid-search shift under a 2 mm early-position constraint is 0.1052 m. Among 100,000 joint perturbations with fixed mass and radius, 95.772% clear the goal. A same-state curved-shot calculation quantifies rotational damping, while relative airflow describes wind effects.

Keywords: Soccer ball, aerodynamic drag, Magnus effect, inverse prediction, trajectory comparison

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1 Introduction

1.1 Problem Background

A penalty kick places stringent spatial and temporal demands on the goalkeeper. The goalkeeper must reach an appropriate location at the right time, while deciding when and in which direction to move. Studies of penalty anticipation discuss this timing problem and show that shot speed and kicking technique affect the predictability of shot direction [1, 2]. Such work also considers cues from the kicker before contact; the present problem instead focuses on predicting the ball's later position from its measured positions after launch.

A kicked soccer ball can curve in flight: gravity pulls it downward, drag slows it, and rotation produces lateral or vertical force. Previous studies combine aerodynamic forces with motion equations and measured trajectories[5], and examine how drag, spin and wind affect three-dimensional flight[10, 8].

The ball's surface also affects its aerodynamic forces. Wind-tunnel measurements relate flow conditions to those forces[11], while simulations compare a seamed football with a smooth sphere[9]. These studies support a spherical approximation followed by an examination of force coefficients.

Measured flights provide a basis for comparison. Ball markings and cameras can track position and spin[3], while trajectories can be used to estimate drag and lift coefficients[15, 4].

The problem supplies an initial position and two later positions, but not initial velocity or spin. We find states compatible with those observations and calculate their goal-plane crossings. Recorded match flights provide a comparison; Appendix A gives the source and processing.

1.2 Problem Restatement

We study a ball travelling from the penalty mark to a goal 7.32 m wide and 2.44 m high, 11 m away. The motion and prediction problem can be organized into three stages:

1. **Ball–foot contact.** Contact establishes launch velocity and spin. These initial conditions relate to the unknown motion reconstructed in Section 4.

2. **Airborne flight.** Gravity, drag and Magnus lift develop the trajectory according to the governing equations established in Section 3.

3. **Goal arrival and shot outcome.** The goal-plane crossing connects the flight to the shot result. Variation in initial motion, aerodynamic parameters and environmental conditions can produce a distribution of crossing locations; Section 5 examines these sensitivities and joint perturbations.

We estimate initial states from three real penalties and compare placement and power shots with later recorded positions. Figure 1 summarizes this physical and computational sequence.

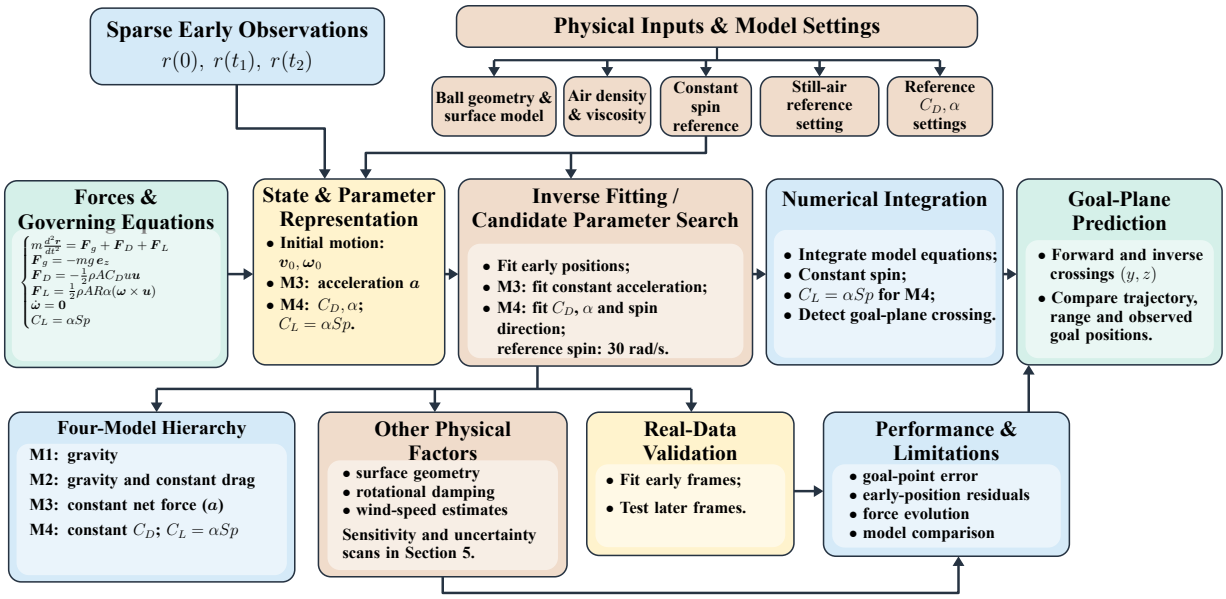


Figure 1: Three-force modeling, inverse prediction and real-penalty validation.

2 Preparation of the Models

2.1 Assumptions

- **Contact and model start.** Real inversions define $t = 0$ at the selected starting tracking frame; contact is represented through the initial motion.
- **Rigid spherical ball.** The ball is idealized as a rigid sphere with constant mass, radius and cross-sectional area $A = \pi R^2$; its centre of mass coincides with its geometric centre [8, 5, 10]. Empirical football coefficients represent the average aerodynamic effect of its surface. Section 6.1 discusses seams and asymmetric geometry.
- **Free-flight motion.** Gravity, aerodynamic drag and spin-induced lift act during free flight [8, 5]. The flight considered ends at the first contact with the ground, goal structure or another object; collisions, bouncing and rolling are excluded. Integration also stops at the selected goal plane.
- **Uniform gravity.** Gravitational acceleration is constant, with $g = 9.81\,\mathrm{m/s^2}$.
- **Atmospheric properties.** Air density and dynamic viscosity are spatially uniform during each flight. Aerodynamic forces are represented using empirical drag and lift coefficients.
- **Wind.** The main calculation uses still air. When wind is included, its velocity vector is spatially and temporally constant during each flight; small-scale turbulence and rapidly varying gusts are neglected. Section 6.3 examines how different wind conditions affect relative airflow and force magnitude.

- **Spin during flight.** The main model neglects aerodynamic torque, so its spin vector remains constant. Section 6.2 examines this approximation by adding rotational damping to the same initial state.
- **Observations and unknowns.** Three specified real-penalty positions enter a bounded deterministic fit, and subsequent positions supply the comparison. Released Z values are interpreted as metres and centre height. Section 4 states the search ranges for velocity, spin and force corrections; Section 5 uses the competition-position example to examine parameter effects on crossing distributions.

2.2 Notations

Table 1: Principal quantities and competition reference values.

Symbol	Meaning	Value or unit
$\boldsymbol{r}, \boldsymbol{v}$	Ball-centre position and velocity	m; m/s
$\boldsymbol{\omega}, \omega$	Spin vector and its magnitude	rad/s
m, R, A	Mass, radius and frontal area	0.43 kg; 0.11 m; πR^2
g	Gravitational acceleration	9.81 m/s^2
$\boldsymbol{u}, u$	Air-relative velocity and its magnitude	m/s
ρ	Air density	1.2 kg/m^3
$\nu, \mu = \rho\nu$	Kinematic and dynamic viscosity	1.50×10^{-5} m^2/s; Pa s
Re, Sp	Reynolds number and spin parameter	dimensionless
C_D, C_L, α	Drag, lift and linear-spin parameters	dimensionless
$\lambda_D, \lambda_\alpha$	Coefficient ratios to reference values	dimensionless
$\theta, \boldsymbol{e}_L$	Spin–airflow angle and Magnus direction	rad; unit vector
X_g	Local goal-plane coordinate	11 m
$\boldsymbol{Q}, I_b$	Damping torque and moment of inertia	N m; kg m^2

The ball and air parameters use published reference values [8, 11]. The viscosity is a declared reference air setting, with $\mu = \rho\nu$. The three match penalties use the same declared reference settings.

3 Model Establishment

The four models describe gravitational fall, drag-induced slowing, a constant-force approximation and aerodynamic curvature. We establish the goal geometry and M4 forces below; Section 4 gives the inverse procedures.

3.1 Geometry and Goal Plane

The local horizontal origin is the nominal penalty mark, with x towards the goal, y lateral and z upward. The goal plane is $x = 11$ m and the opening is $-3.66 \le y \le 3.66$ m, $0 \le z \le 2.44$ m.

Section 4.1 converts the measured pitch-centred coordinates.

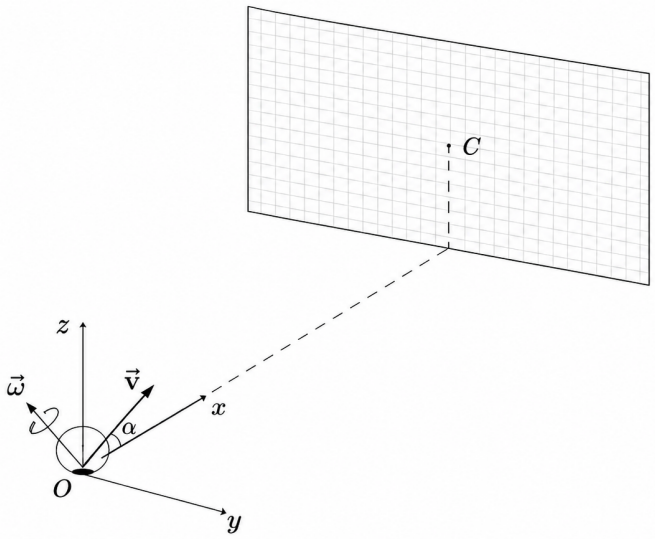


Figure 2: Penalty-kick geometry in local coordinates; O is the nominal penalty mark and C an illustrative goal-plane point.

A clean opening crossing requires $|y| \leq 3.66 - R$ and $R \leq z \leq 2.44 - R$. Descending centre height $z = R$ defines ground contact. Every model uses these same event definitions.

3.2 Force Analysis

3.2.1 Gravity

Gravity supplies a constant downward force,

$$\boldsymbol{F}_g = -mg\boldsymbol{e}_z, \tag{1}$$

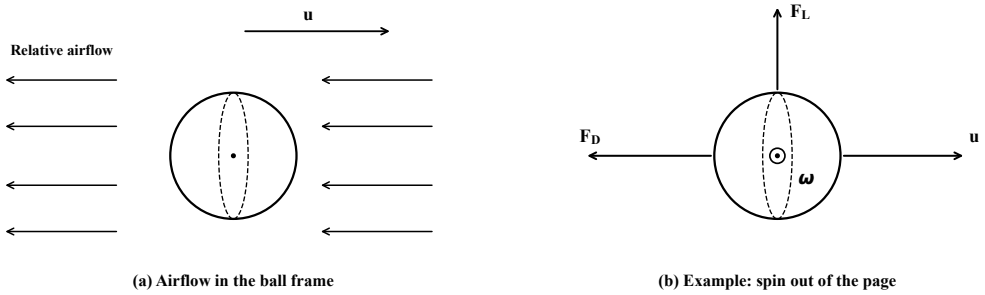
where $\boldsymbol{e}_z$ is the upward vertical unit vector. It produces downward curvature even when all aerodynamic forces are removed.

By Archimedes' principle the displaced air exerts an upward force on the ball. Its volume is $V_b = \frac{4}{3}\pi R^3$, giving

$$\boldsymbol{F}_B = \rho V_b g\, \boldsymbol{e}_z = \frac{4}{3}\pi R^3 \rho g\, \boldsymbol{e}_z. \tag{2}$$

Together, gravity and buoyancy give $\boldsymbol{F}_g + \boldsymbol{F}_B = -(m - \rho V_b)g\boldsymbol{e}_z$. Buoyancy is 1.56% of weight for the reference parameters. The baseline calculation uses gravity, drag and Magnus force, treating buoyancy as a small vertical correction.

Figure 3 illustrates the relative airflow in the ball frame and the corresponding force directions. The right-hand example takes the spin axis out of the page; the lift direction follows the cross product.



(a) Airflow in the ball frame (b) Example: spin out of the page

Figure 3: Relative airflow and force directions. The symbol $\odot$ denotes spin out of the page; $\boldsymbol{u}$ in the left panel is the velocity of the ball relative to the air.

3.2.2 Aerodynamic Drag

With relative velocity $\boldsymbol{u} = \boldsymbol{v} - \boldsymbol{v}_{\text{wind}}$, $u = \|\boldsymbol{u}\|$ and $A = \pi R^2$, drag is [15, 12]

$$\boldsymbol{F}_D = -\tfrac{1}{2}\rho A C_D u \boldsymbol{u}. \tag{3}$$

The drag coefficient C_D is constant during each flight.

3.2.3 Magnus Lift

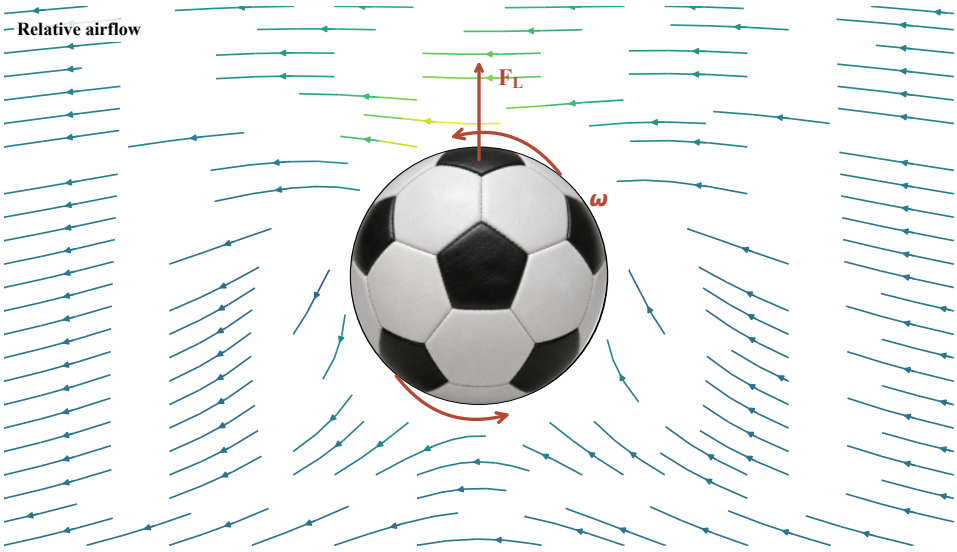


Figure 4: Idealized flow around a spinning football.

Let $Re = 2Ru/\nu$, $Sp = R\|\boldsymbol{\omega}\|/u$ and $C_L = \alpha Sp$. Resolving the transverse component yields

$$\boldsymbol{F}_L = \tfrac{1}{2}\rho A R \alpha (\boldsymbol{\omega} \times \boldsymbol{u}). \tag{4}$$

The force vanishes at zero spin or zero relative velocity. The forces on the football are summarized in the following diagram.

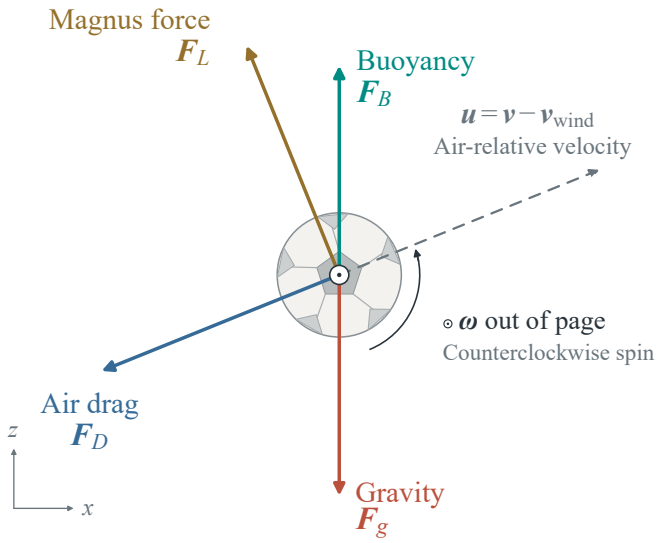


Figure 5: Forces on the football: gravity, buoyancy, drag and Magnus force, with arrows showing their directions for the illustrated state.

3.3 Dynamical Equations

Newton's second law and angular-momentum balance give

$$m\dot{\boldsymbol{v}} = \sum \boldsymbol{F}, \qquad \frac{d\boldsymbol{L}}{dt} = \boldsymbol{Q}, \qquad \boldsymbol{L} = I_b\boldsymbol{\omega}. \tag{5}$$

With zero applied torque and constant spherical inertia, the main model is

$$\begin{cases} \dot{\boldsymbol{r}} = \boldsymbol{v}, \\ m\dot{\boldsymbol{v}} = \boldsymbol{F}_g + \boldsymbol{F}_D + \boldsymbol{F}_L, \\ \dot{\boldsymbol{\omega}} = \boldsymbol{0}. \end{cases} \tag{6}$$

Its initial conditions are

$$\begin{cases} \boldsymbol{r}(0) = \boldsymbol{r}_0, \\ \boldsymbol{v}(0) = (v_{x0}, v_{y0}, v_{z0}), \\ \boldsymbol{\omega}(0) = (\omega_{x0}, \omega_{y0}, \omega_{z0}). \end{cases} \tag{7}$$

The observed initial position fixes the origin of the flight. The six velocity and spin components determine its initial motion. In still air, drag removes mechanical energy while Magnus lift does no work along the velocity; gravity converts kinetic and potential energy during flight.

4 Model Solving

Early positions from three actual penalties determine inverse states whose continuations are compared with later observations. Two placement shots and one power shot provide different speeds, curvature directions and flight heights.

Table 2: Four-model hierarchy and inverse quantities.

Model	Force or motion assumption	Inverse quantities
M1	gravity	initial velocity (3)
M2	gravity and drag	velocity and drag coefficient (4)
M3	constant resultant force	velocity and acceleration (6)
M4	gravity, drag and Magnus force	velocity, spin direction, C_D, α (7)

The M3 resultant includes gravity and the other forces together. In M4, C_D and α remain constant during a flight, while $C_L = \alpha Sp$ changes with speed. The references $C_D^* = 0.2415$ and $\alpha^* = 1.5251$ use $Re^* = 333793$ and $Sp^* = 0.19$, with $\alpha^* = C_L(Re^*, Sp^*)/Sp^*$ [12].

4.1 Inverse Prediction from Three Real Penalties

4.1.1 Observations and numerical method

The first selected position defines the time origin; the three inputs occur at $t = 0, 0.04, 0.08$ s. The penalty mark is the horizontal origin, with x towards the goal, y lateral and z the centre height. The goal plane is $x = 11$ m. Data sources and source frames appear in Appendix A.

Integration uses DOP853 with a maximum step of 0.005 s, relative tolerance 10^{-10} and absolute tolerance 10^{-12}. Forward roots of $x - 11 = 0$ and descending roots of $z - R = 0$ locate the goal crossing and first ground contact.

4.1.2 Inversion and reference states

M1 fits velocity, while M2 fits velocity and drag coefficient. M3 solves the three-position relation

$$\boldsymbol{r}(t) = \boldsymbol{r}_0 + \boldsymbol{v}_0 t + \tfrac{1}{2}\boldsymbol{a}t^2. \tag{8}$$

M4 fits three velocity components, two spin-direction angles and two coefficients by minimizing squared coordinate errors at the two later positions. Six deterministic starts are optimized and the smallest-residual result is selected; Table 11 gives the ranges. Since $\alpha\boldsymbol{\omega}$ determines rotational deflection, we use a 30 rad/s reference magnitude and fit direction and α.

Subsequent path and goal-plane errors are

$$\begin{aligned} \mathrm{RMSE}_{3D} &= \sqrt{\frac{1}{n}\sum_{i=1}^{n} \|\widehat{\boldsymbol{r}}(t_i) - \boldsymbol{r}_i\|^2}, \\ E_g &= \sqrt{(\widehat{y}_g - y_g)^2 + (\widehat{z}_g - z_g)^2}. \end{aligned} \tag{9}$$

Path comparisons use observations after the fitted times and before first contact or the goal crossing, with sample count n listed in the table. E_g is evaluated when a model crosses the goal plane; its observed counterpart comes from a source frame or interpolation between frames.

4.2 Case Analysis and Validation

Figure 6 illustrates the two shot styles.



(a) Illustrative placement shot.



(b) Illustrative power shot.

Figure 6: Photographic examples of placement and power shots.

Figure 7 summarizes the spatial paths. Open circles are inverse inputs, black points and lines are observations, coloured lines are predictions, and crosses are observed goal crossings.

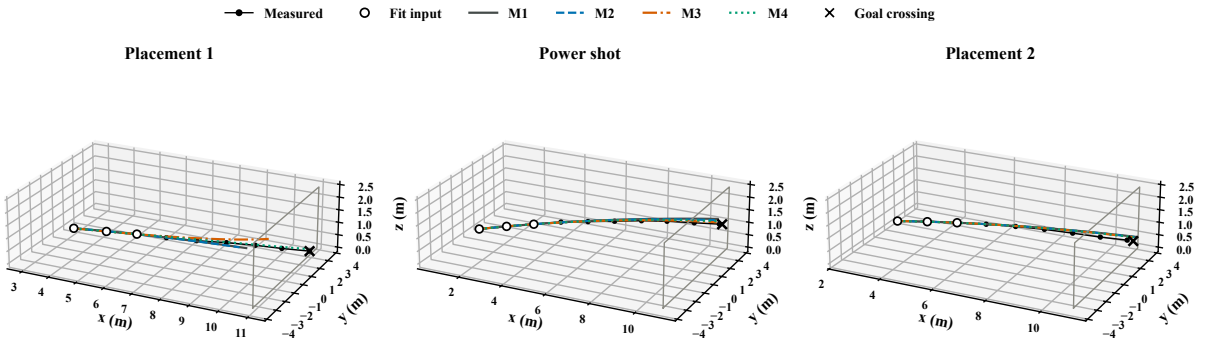


Figure 7: Observed and predicted paths for placement 1, the power shot and placement 2, from left to right. Fitting times are 0, 0.04 and 0.08 s in each case.

4.2.1 Case 1: Placement shots

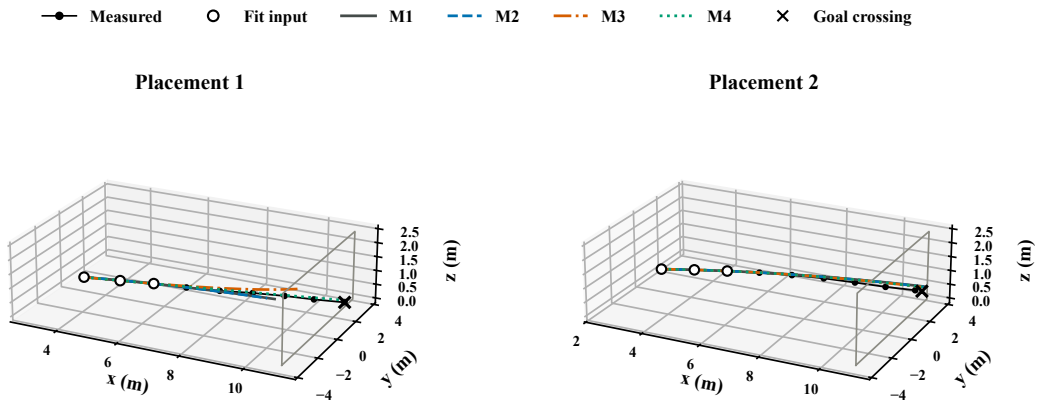


Figure 8: Measured and predicted trajectories for the two placement cases.

Both placement shots travel close to the ground. M4 gives starting speeds of 29.20 m/s and 28.34 m/s. For placement 1, M1 and M2 reach the ground before the goal. M3 gives $a_x = -56.25$ m/s^2; Eq. (8) then gives a maximum $x = 10.973$ m, short of the goal. M4 reaches the plane with a crossing error of 0.360 m and a subsequent path RMSE of 0.234 m.

All four models reach the goal for placement 2, with crossing errors 0.475, 0.475, 0.155 and 0.146 m. Constant acceleration carries forward the early curvature in this case; M4 reduces the path RMSE from M3's 0.151 m to 0.132 m. Gravity draws a low flight downward, while the vertical Magnus component raises or lowers it according to spin direction. Together they determine whether the ball reaches the goal before contact.

4.2.2 Case 2: Power shot

M4 gives a starting speed of 32.05 m/s. Figure 9 displays the spatial path and coordinate histories.

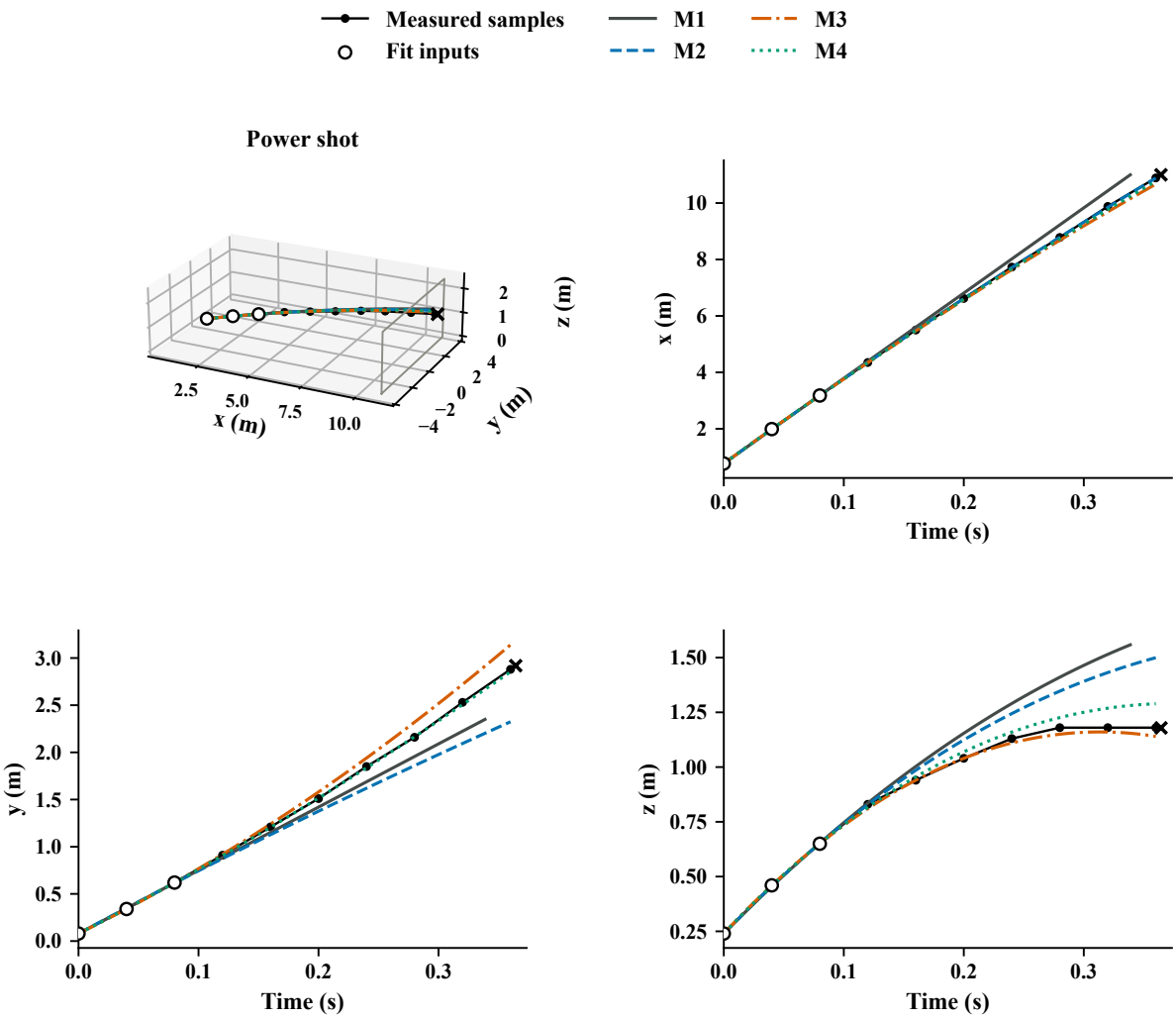


Figure 9: Power-shot path and coordinate histories. Open circles are inverse inputs, black points are observations, and coloured curves are predictions.

M1–M4 have crossing errors of 0.684, 0.657, 0.374 and 0.113 m, and subsequent path RMSE values of 0.412, 0.356, 0.190 and 0.093 m. Lateral motion in M1 and M2 follows the initial velocity; M3 extends constant acceleration. M4 curves the path through changing drag and Magnus forces, giving the smallest values of both errors for this shot.

Speed and aerodynamic forces. Figure 10 gives the M4 speed and force histories. Initial drag is 7.58 N for the power shot and 7.05 N and 2.21 N for the placement shots. Drag depends on $C_D v^2$, combining speed with the fitted coefficient, and decreases as the flight slows. Linear lift depends on $\alpha(\boldsymbol{\omega} \times \boldsymbol{v})$; the angle and direction between spin and velocity determine its lateral and vertical effects.

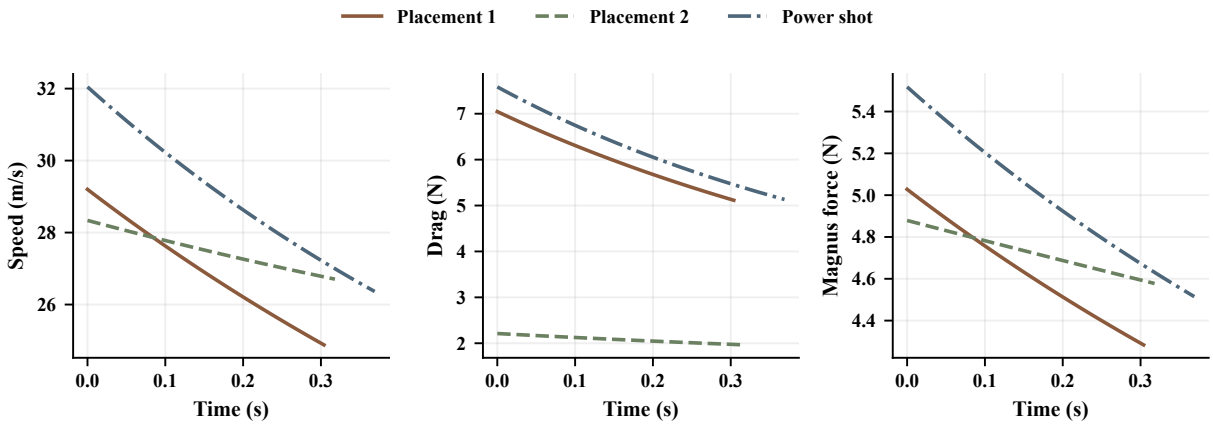


Figure 10: M4 speeds, drag and Magnus force for the three penalties. Curves end at first ground contact or the goal plane.

Table 3: Early maximum coordinate residual, subsequent path RMSE/sample count, goal-plane error and termination.

Event	Model	Residual (mm)	RMSE (m)/n	Goal (m)	Status
Placement 1	M1	36.00	0.244/3	—	ground
Placement 1	M2	26.13	0.136/3	—	ground
Placement 1	M3	0.00	0.826/6	—	no crossing
Placement 1	M4	25.90	0.234/5	0.360	goal
Power shot	M1	12.00	0.412/6	0.684	goal
Power shot	M2	10.28	0.356/7	0.657	goal
Power shot	M3	0.00	0.190/7	0.374	goal
Power shot	M4	2.90	0.093/7	0.113	goal
Placement 2	M1	8.00	0.300/5	0.475	goal
Placement 2	M2	7.01	0.320/6	0.475	goal
Placement 2	M3	0.00	0.151/5	0.155	goal
Placement 2	M4	1.12	0.132/5	0.146	goal

5 Sensitivity Analysis

The competition's three positions provide a reference example for examining initial motion, aerodynamic coefficients and air conditions. Mass and radius remain fixed at $m = 0.43$ kg and $R = 0.11$ m, with the M4 forces of Section 3.

Table 4: Early positions for the sensitivity example.

t (s)	x (m)	y (m)	z (m)
0	0	0	0.11
0.05	1.50	0.45	0.44
0.10	2.96	0.86	0.74

Using the 30 rad/s reference spin magnitude of Section 4 gives

$$\begin{aligned} \boldsymbol{v}_0 &= (30.3784, 9.4074, 6.8974)\ \mathrm{m/s}, \\ \boldsymbol{\omega}_0 &= (19.7434, 1.2060, -22.5553)\ \mathrm{rad/s}. \end{aligned} \tag{10}$$

Here $C_D = 0.3623$ and $\alpha = 2.2877$, with a maximum early-coordinate residual of 0.784 mm. The reference flight crosses the goal plane at $(y_g^*, z_g^*) = (2.6159, 1.9526)$ m and $t_g^* = 0.398967$ s.

5.1 Effects of individual parameters

One parameter changes at a time. Speed varies by $\pm 5\%$; spin magnitude, C_D, α and air density each vary by $\pm 10\%$; velocity and spin-axis azimuth and elevation each vary by $\pm 2^\circ$. Each scan uses 41 equally spaced values, giving 369 flights. Define

$$\begin{aligned} \Delta y_g &= y_g - y_g^*, \quad \Delta z_g = z_g - z_g^*, \quad \Delta t_g = t_g - t_g^*, \\ D_g &= \sqrt{(\Delta y_g)^2 + (\Delta z_g)^2}. \end{aligned} \tag{11}$$

Table 5 compares direction changes of $\pm 1^\circ$ and other changes of $\pm 5\%$, taking the larger D_g and $|\Delta t_g|$ from each positive–negative pair.

Table 5: Goal-plane displacement and arrival-time change under one-factor perturbations.

Parameter	Perturbation	max D_g (cm)	max $\lvert\Delta t_g\rvert$ (ms)
Launch speed	$\pm 5\%$	8.356	20.659
Velocity azimuth	$\pm 1^\circ$	20.070	1.965
Velocity elevation	$\pm 1^\circ$	20.218	1.692
Spin magnitude	$\pm 5\%$	3.873	0.287
Spin-axis azimuth	$\pm 1^\circ$	1.002	0.071
Spin-axis elevation	$\pm 1^\circ$	0.852	0.084
Drag coefficient	$\pm 5\%$	0.620	2.274
Lift parameter	$\pm 5\%$	3.873	0.287
Air density	$\pm 5\%$	4.142	1.990

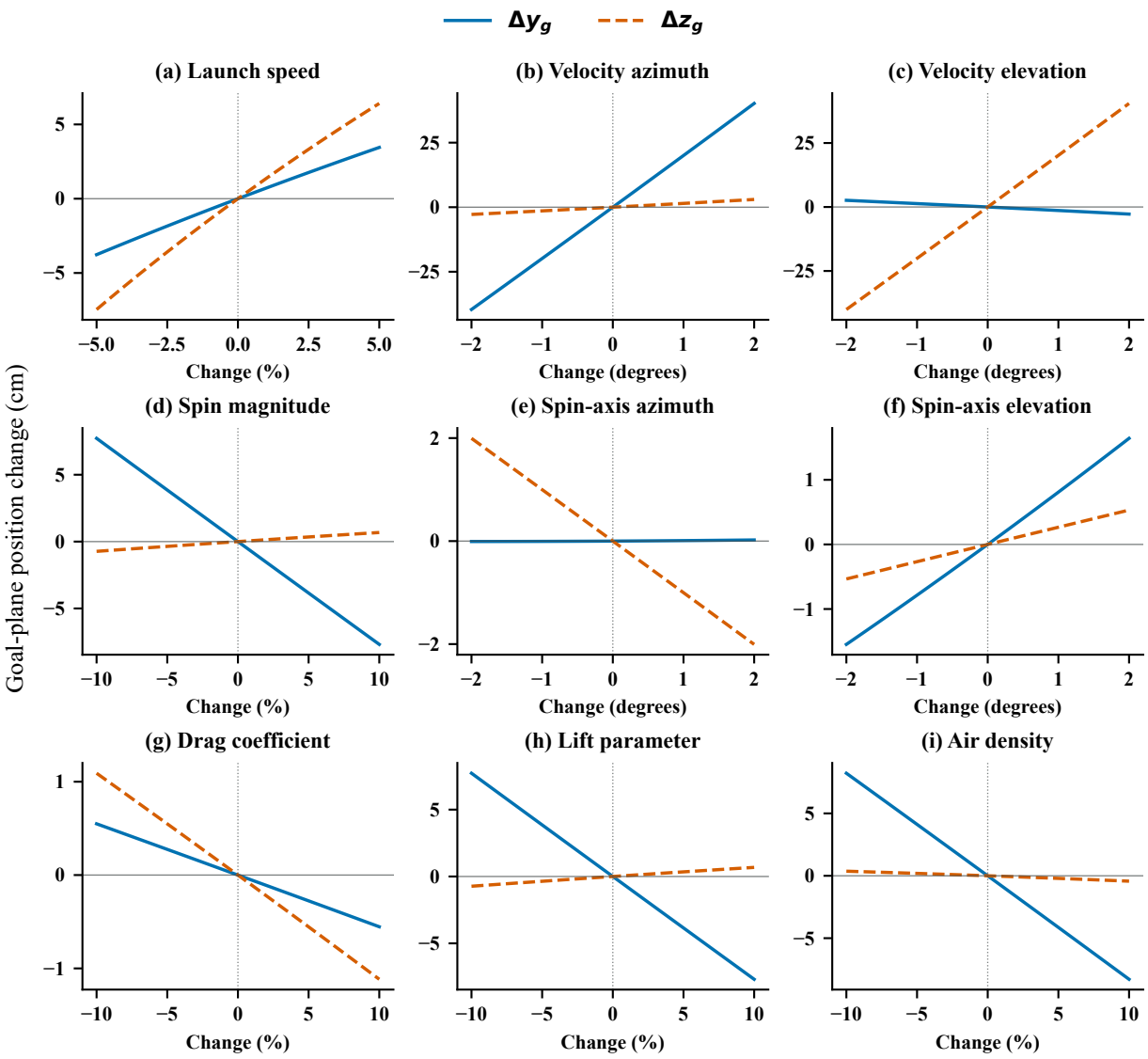


Figure 11: Goal-plane coordinate responses to nine one-factor perturbations. Blue solid curves show Δy_g and orange dashed curves Δz_g, in cm. Vertical scales differ between panels.

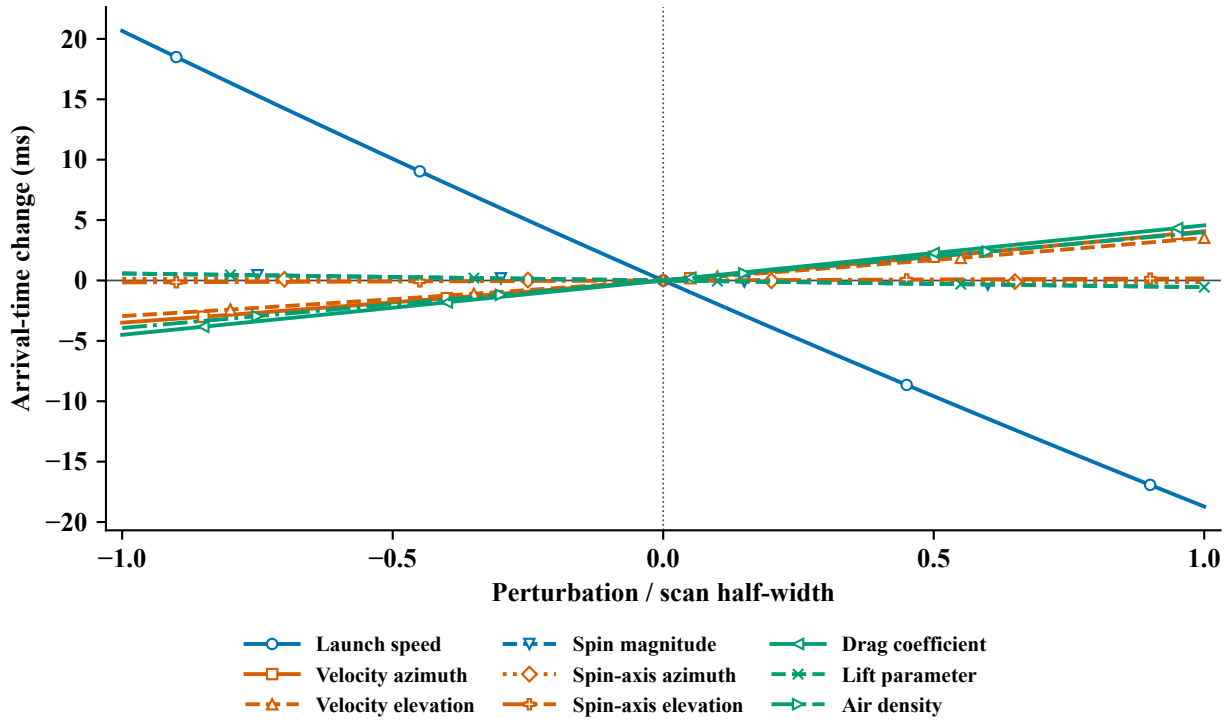


Figure 12: One-factor arrival-time responses. Perturbation divided by scan half-width is plotted horizontally, so ± 1 marks each scan endpoint. The vertical axis is Δt_g in ms; positive values mean later arrival.

A 1° change in velocity azimuth or elevation gives a maximum displacement of 20.070 cm or 20.218 cm. A 5% speed change gives 8.356 cm and 20.659 ms. Speed sets the time available to reach the goal, while direction directly changes lateral and vertical displacement; drag further alters flight time through deceleration.

The linear lift law contains the product of α and spin magnitude. Applying the same relative change to either quantity therefore produces coincident response curves. For any $k > 0$,

$$\boldsymbol{\omega}_0' = k\boldsymbol{\omega}_0, \qquad \alpha' = \alpha/k \quad \Longrightarrow \quad \boldsymbol{F}_L' = \boldsymbol{F}_L. \tag{12}$$

This compensation explains the reference spin convention and connects the individual scans with the position-constrained analysis below.

5.2 Crossing variation under early-position constraints

We retain the positions in Table 4 and refit velocity and spin to examine aerodynamic parameter combinations. With the coefficient references C_D^*, α^*, define $\lambda_D = C_D/C_D^*$ and $\lambda_\alpha = \alpha/\alpha^*$. Each multiplier takes 17 values in $[1.2, 1.8]$. At each node, three velocity and three spin components are fitted, using total speed at most 35 m/s, spin components at most 60 rad/s in magnitude, and maximum early-coordinate residual at most 2 mm.

The grid produces 153 compatible flights; 63 lie in the inner square $[1.35, 1.65]^2$. Table 6 gives crossings and arrival times. The largest crossing displacement in the wider scan is 0.1052 m.

Table 6: Ranges of goal-plane crossings compatible with the early positions.

Range of each multiplier	y_g (m)	z_g (m)	t_g (s)
$[1.35, 1.65]$	2.5538–2.6550	1.9122–2.0307	0.3973–0.4025
$[1.2, 1.8]$	2.5478–2.6601	1.9075–2.0368	0.3973–0.4043

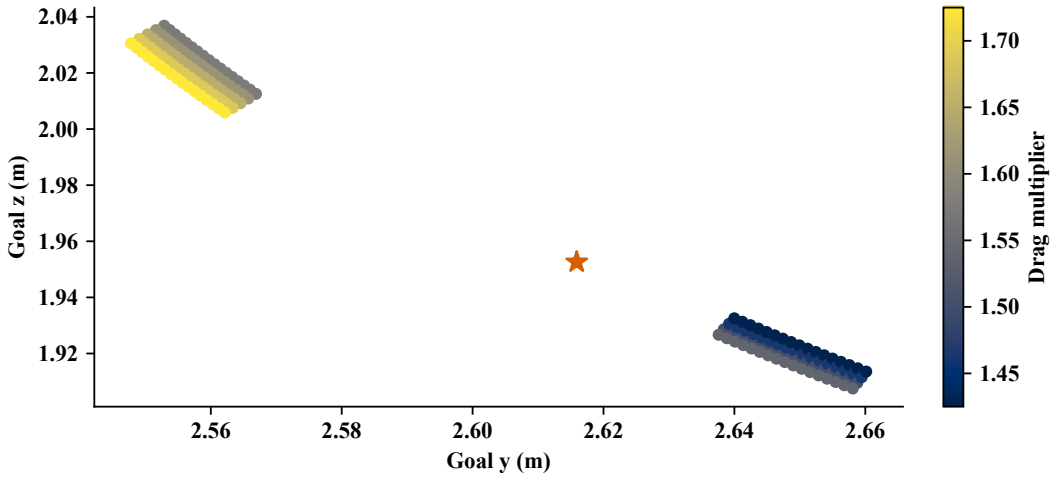


Figure 13: Grid-search crossings under the 2 mm early-position constraint. Colour indicates λ_D; the orange star is the reference flight.

Different forces and initial states can stay close at early times and separate later, producing the lateral and vertical spread shown here. Refitting the other inverse quantities connects this spread to compensation under the position constraints.

5.3 Joint perturbations and Monte Carlo simulation

Nine parameters are sampled independently and uniformly over the full ranges in Section 5.1. Seed 20261108 generates 100000 trajectories. Mass, radius and initial position remain fixed while launch motion and aerodynamic coefficients vary together. Trajectories use fourth-order Runge–Kutta with a 0.002 s step and cubic interpolation of the crossing event. Every trajectory reaches the goal plane before ground contact.

Table 7: Monte Carlo crossing positions and arrival times (100,000 trajectories).

Output	Sample minimum–maximum	Central 95% interval, one coordinate
y_g (m)	1.9633–3.2143	2.1858–3.0505
z_g (m)	1.4381–2.4646	1.5534–2.3535
t_g (s)	0.3685–0.4346	0.3783–0.4218

The central 95% intervals are the 2.5th–97.5th percentiles of each output. The mean crossing

is $(2.6170, 1.9511)$ m, with lateral and vertical standard deviations 0.2455 m and 0.2360 m. The largest displacement from the reference crossing is 0.801 m.

For a joint region, let $\boldsymbol{q}_i = (y_{g,i}, z_{g,i})^T$, with sample mean $\bar{\boldsymbol{q}}$ and covariance $\widehat{\Sigma}$. Use

$$\begin{aligned} d_i^2 &= (\boldsymbol{q}_i - \bar{\boldsymbol{q}})^T \widehat{\Sigma}^{-1} (\boldsymbol{q}_i - \bar{\boldsymbol{q}}), \\ \mathcal{E}_{0.95} &= \{\boldsymbol{q} : (\boldsymbol{q} - \bar{\boldsymbol{q}})^T \widehat{\Sigma}^{-1} (\boldsymbol{q} - \bar{\boldsymbol{q}}) \le c_{0.95}\}. \end{aligned} \tag{13}$$

The empirical 95th percentile $c_{0.95} = 4.5916$ gives an ellipse containing 95% of the simulated crossings.

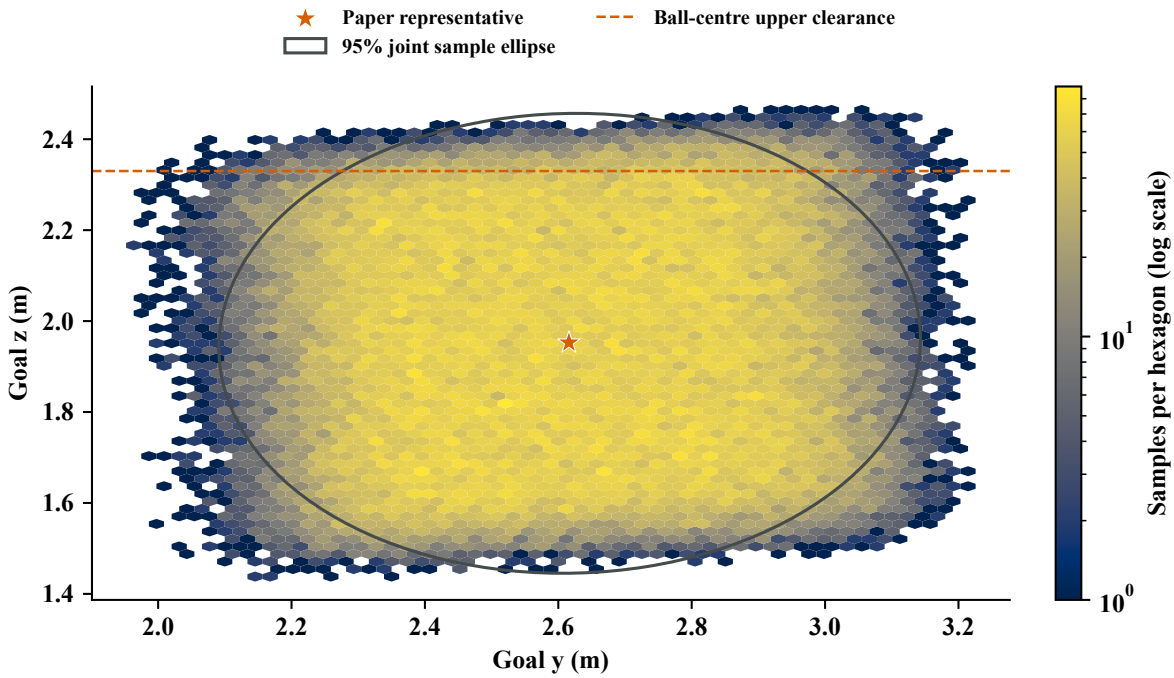


Figure 14: Goal-plane crossings under 100,000 joint perturbations. Hexagon colour indicates sample count, the ellipse gives the joint 95% region, and the dashed line marks the centre-height clearance limit $z_g = 2.33$ m.

The clearance rule of Section 3 gives 95772 flights (95.772%) passing fully through the opening and 4228 (4.228%) crossing too high.

Figure 15 displays 400 randomly selected flights. Each line connects nine inputs to y_g, z_g, t_g. Inputs use their perturbation limits and outputs the full-sample extrema for linear scaling to 0–1. Blue lines clear the opening, orange lines are too high, and the grey dashed line is the reference state.

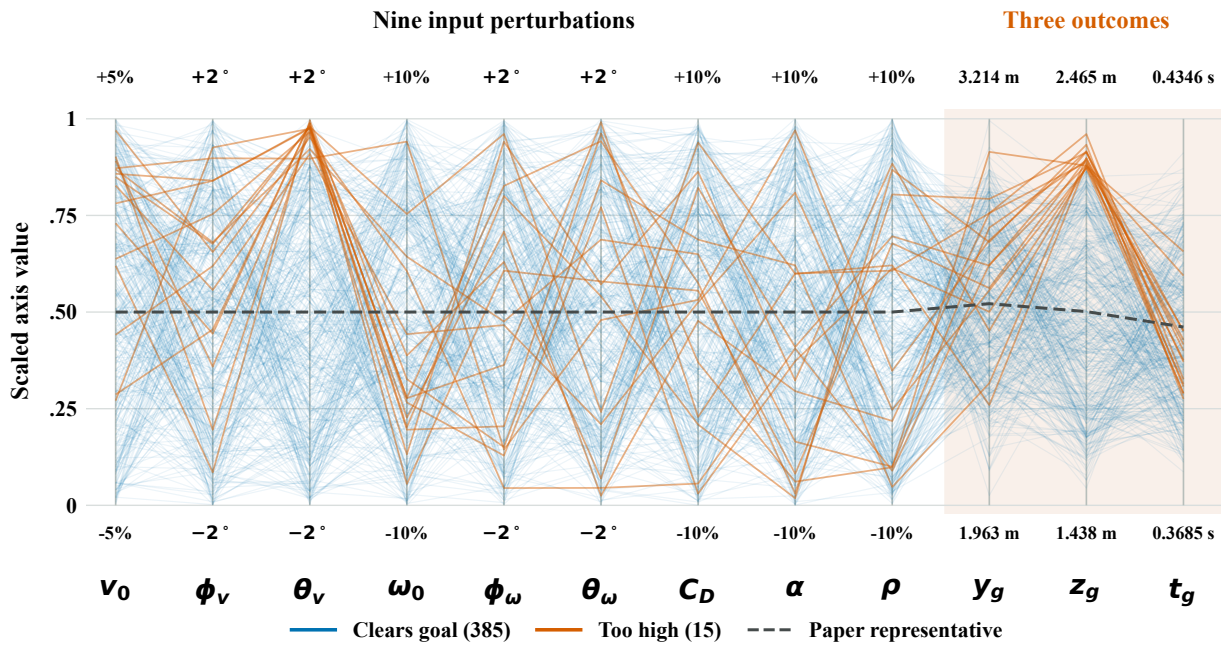


Figure 15: Parallel coordinates linking inputs, goal-plane positions and arrival time. The first nine axes are inputs; three shaded axes are outputs. The displayed subset contains 385 clear flights and 15 high flights.

Following a line reveals the elevation, spin, coefficients and arrival time associated with a high crossing. The spread on the output axes shows lateral and vertical changes together.

Increasing the sample from 50000 to 100000 changes the central-95% interval endpoints by at most 0.264 mm laterally, 0.506 mm vertically and 0.138 ms in time. Two independent seeds each produce 10000 flights. Their largest interval-endpoint differences from the full sample are 5.425 mm laterally, 4.320 mm vertically and 0.328 ms in time.

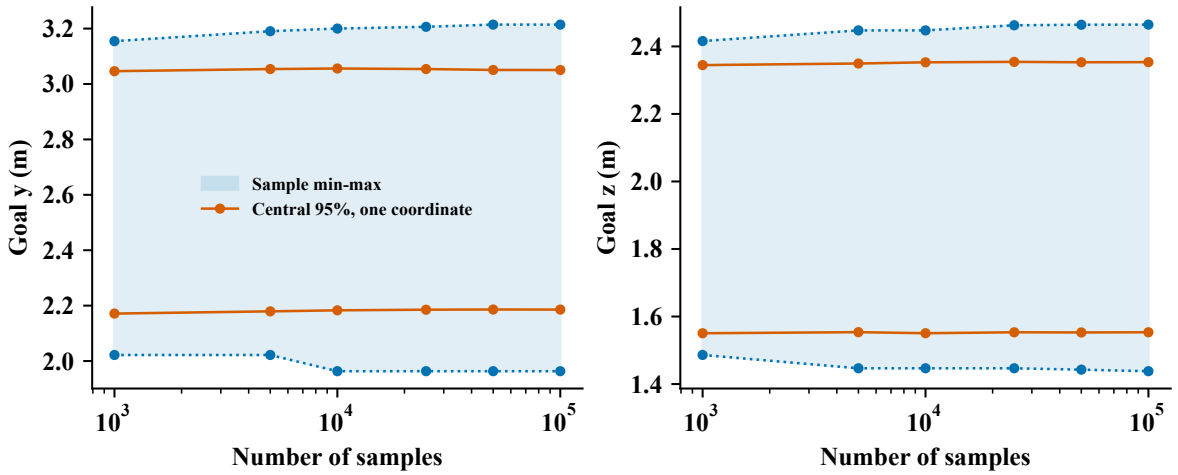


Figure 16: Sample-count effects on crossing ranges and percentile intervals. Blue shading and dotted curves show sampled extrema; orange curves show each coordinate's central 95% interval.

6 Discussion of Other Factors

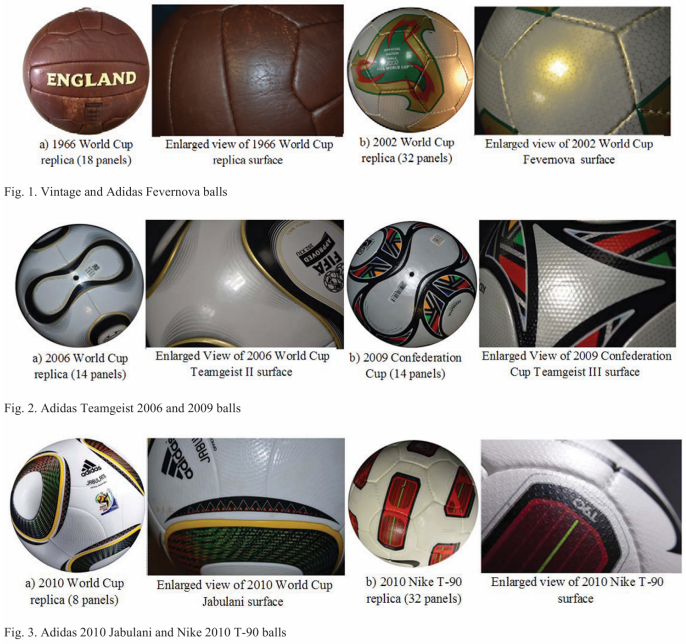
The main model isolates gravity, drag and Magnus lift. We now examine the physical effects omitted from its geometry, spin evolution and ambient airflow.

6.1 Asymmetric Surface Geometry

A full-flight simulation compares realistic football geometry with a smooth sphere [9]. For launch speeds of 10, 15 and 20 m/s and spin rates of 31.5, 63 and 94.5 rad/s, their results show that seams and surface structure alter separation, wake dynamics and Magnus force. The smooth sphere produces greater lateral bending in their nine comparisons, and vertical flight-time differences become more visible at higher speed and rotation.



(a) Illustrative seamed-football and smooth-sphere comparison [9].



(b) Six footballs and surface details, reproduced from Figures 1–3 of Alam et al. [16].

Figure 17: Football geometry, panels and surface textures.

Wind-tunnel experiments measured drag coefficients over a range of Reynolds numbers for six footballs and a smooth sphere [16]. Figure 17b shows the panels and surface textures of these footballs, while Figure 18 compares their drag coefficients. The differences illustrate the influence of ball surface structure on aerodynamic drag.

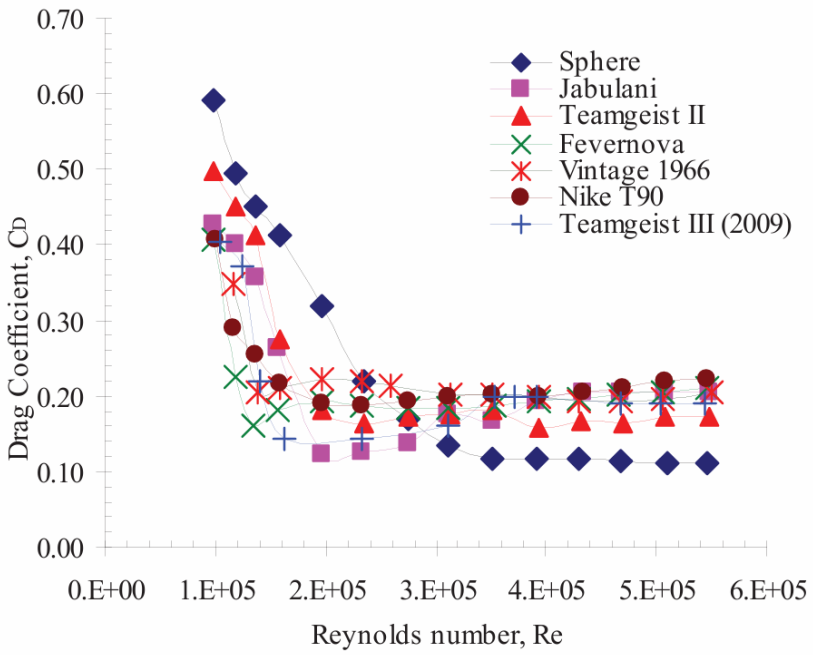


Figure 18: Drag coefficient against Reynolds number for different footballs and a smooth sphere, reproduced from Figure 5 of the source[16].

The spherical geometry in the present report describes the centre-of-mass motion and projected area. Its force coefficients come from football measurements, so the average surface influence enters through C_D and C_L. Explicit panel orientation and asymmetric wake fluctuations provide a further level of description when lateral-force details are important. These published results support retaining football-specific coefficients in a spherical trajectory model.

6.2 Rotation and Aerodynamic Damping

Aerodynamic torque opposes spin, gradually weakening rotation and changing the Magnus force. We use the high-Reynolds-number rotational-damping model described by Grad et al. and MuJoCo [13, 14], in which stronger spin produces stronger resistance. The angular-drag parameter is the software reference value $C_{\text{angular}} = 1.5$, with the thin-shell inertia approximation [8]. Viscous damping is much weaker than the inertial contribution in this example, so the comparison focuses on the latter.

The reference curved shot has initial velocity $(23.6354, 0, 4.1676)$ m/s, spin $(0, 0, 35)$ rad/s and unit force corrections. Holding this state fixed, we and compare constant spin with rotational damping. Figure 19 shows both the spin evolution and the resulting trajectory difference.

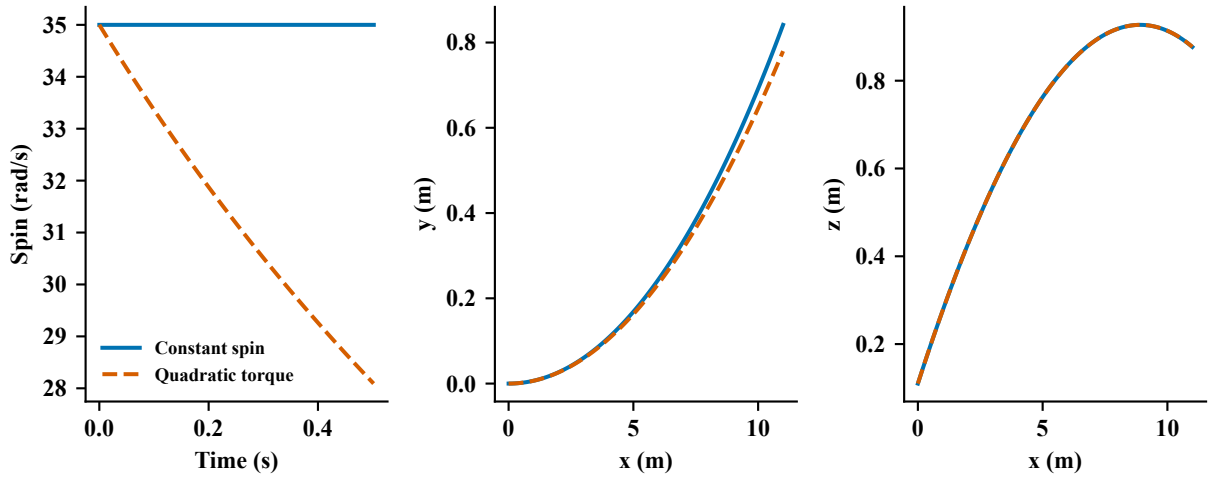


Figure 19: Constant spin and rotational damping under the same initial curved-shot conditions.

Spin falls by 19.740%, while the crossing moves 0.0616 m (0.560% of the 11 m distance) and arrival time changes by -0.066%. Rotational damping therefore has a small endpoint effect in this short reference flight.

6.3 Wind Speed and Force Estimates

Wind changes the airflow through $\boldsymbol{u} = \boldsymbol{v} - \boldsymbol{v}_{\text{wind}}$ [8]. At fixed coefficients, a wind speed w along the flight direction changes the dynamic-pressure scale by

$$\frac{q_{\text{head}}}{q_0} = (1 + w/v)^2, \qquad \frac{q_{\text{tail}}}{q_0} = (1 - w/v)^2, \qquad \frac{q_{\text{cross}}}{q_0} = 1 + (w/v)^2. \tag{14}$$

Crosswind rotates the airflow by $\arctan(w/v)$ and changes the force direction. Table 8 evaluates 1, 3 and 5 m/s winds at the 24 m/s curved-shot speed. For the M4 linear lift law, Eq. (4) also gives the headwind and tailwind effects along the flight direction. With fixed spin and $0 \le w < v$,

$$\frac{F_{L,\text{head}}}{F_{L,0}} = 1 + \frac{w}{v}, \qquad \frac{F_{L,\text{tail}}}{F_{L,0}} = 1 - \frac{w}{v}. \tag{15}$$

Table 8: Illustrative wind at the 24 m/s curved-shot launch: dynamic-pressure changes at fixed coefficients.

w (m/s)	Headwind (%)	Tailwind (%)	Crosswind (%)	Angle (°)
1	+8.51	-8.16	+0.17	2.39
3	+26.56	-23.44	+1.56	7.13
5	+46.01	-37.33	+4.34	11.77

A 5% limit on the headwind dynamic-pressure change corresponds to $w/v \le \sqrt{1.05} - 1$, or $w \le 0.59$ m/s at this launch. The still-air approximation is most suitable for indoor or weak-wind flight; stronger wind should enter through the relative-velocity calculation.

7 Conclusions

The four models describe gravitational fall, deceleration, a constant resultant force and rotational deflection. M3 obtains constant acceleration from three positions. M4 inserts constant C_D and $C_L = \alpha Sp$ into the motion equations, updating drag and Magnus force as speed changes.

M4 gives crossing errors of 0.360, 0.146 and 0.113 m for the three penalties. Its power-shot crossing error and subsequent path RMSE are the smallest among the four models. For placement 1, M3 passes through the early positions but reaches a maximum $x = 10.973$ m, short of the goal; M4 supplies a goal-plane prediction for this low flight.

In the competition-position example, a 1° launch-direction change moves the crossing by about 0.20 m, while speed strongly affects arrival time. Spin magnitude and α compensate through the linear lift law. The largest grid-search shift under the 2 mm position constraint is 0.1052 m; 95.772% of 100,000 jointly perturbed flights clear the opening.

Rotational damping shifts the reference curved-shot crossing by 0.0616 m. Wind enters through relative velocity: along the flight direction, drag changes with the squared speed ratio and linear Magnus force with the speed ratio.

8 Model Evaluation

8.1 Model Strengths

Physical interpretation. The model connects the path to three familiar effects: gravity bends it downward, drag reduces speed, and Magnus force redirects a spinning ball. The four-model comparison separates these mechanisms from a constant-acceleration description, making the differences between low placement shots and a higher power shot easier to explain.

A common framework for inference and prediction. Three early positions provide the input for estimating initial motion and continuing the flight with the same governing equations. Speed, force histories and goal-plane crossings can therefore be interpreted together. The common coordinate system and termination rules also make comparisons between models and recorded shots consistent.

Comparison with measurements. Published aerodynamic measurements support the coefficient choices, while later positions from three actual penalties assess the predicted continuation. The error table, trajectory plots and sensitivity study link the physical assumptions to observable differences. The wind and rotational-damping comparisons further identify when the simpler flight assumptions give useful approximations.

8.2 Model Weaknesses

Sensitivity of the inferred state. The fitted velocity, spin direction and aerodynamic parameters depend on the early tracking positions, frame selection and search bounds. Position differences over short time intervals strongly influence estimated acceleration, especially for flights close to the ground. The reference spin magnitude and the coupling between spin and lift parameter also affect the selected M4 state.

Simplified aerodynamics and environment. Constant coefficients during a flight smooth over

flow transitions, panel orientation and fluctuating wakes. The spherical, still-air and constant-spin assumptions likewise average effects that may become more important for stronger winds, sustained spin or longer flight. Their relative importance changes with the ball and operating conditions.

Limited coverage of recorded conditions. The three penalties cover a modest range of shot types, balls and environments. Broader comparison would benefit from ball-specific aerodynamic data, directly measured spin and more early tracking positions. These additions would strengthen the initial-state estimate and help determine when wind and rotational damping should enter the main prediction.

References

[1] J. van der Kamp, M. Dicks, J. A. Navia and B. Noël, "Goalkeeping in the soccer penalty kick," *German Journal of Exercise and Sport Research*, 48, 169–175, 2018. https://doi.org/10.1007/s12662-018-0506-3.

[2] A. H. Hunter, S. C. Murphy, M. J. Angilletta Jr. and R. S. Wilson, "Anticipating the direction of soccer penalty shots depends on the speed and technique of the kick," *Sports*, 6, 73, 2018. https://doi.org/10.3390/sports6030073.

[3] I. Griffiths, C. Evans and N. Griffiths, "Tracking the flight of a spinning football in three dimensions," *Measurement Science and Technology*, 16, 2056–2065, 2005. https://doi.org/10.1088/0957-0233/16/10/022.

[4] J. E. Goff, J. Kelley, C. M. Hobson, K. Seo, T. Asai and S. B. Choppin, "Creating drag and lift curves from soccer trajectories," *European Journal of Physics*, 38, 044003, 2017. https://doi.org/10.1088/1361-6404/aa6fcd.

[5] K. Bray and D. G. Kerwin, "Modelling the flight of a soccer ball in a direct free kick," *Journal of Sports Sciences*, 21, 75–85, 2003. https://doi.org/10.1080/0264041031000070994.

[6] M. Bassek, R. Rein, H. Weber and D. Memmert, "An integrated dataset of spatiotemporal and event data in elite soccer," *Scientific Data*, 12, 195, 2025. https://doi.org/10.1038/s41597-025-04505-y.

[7] DFL, "Sportec Open DFL Dataset," CC BY 4.0. Official release: https://doi.org/10.6084/m9.figshare.28196177. Data mirror: https://huggingface.co/datasets/pysport/idsse-data.

[8] P. Maponi and R. Piombin, "A computational framework to predict the soccer ball trajectory," *Journal of Applied Mathematics*, 2025, 7425581. https://doi.org/10.1155/jama/7425581.

[9] S. Paudel and J. Yan, "High-fidelity simulations of full-flight soccer ball aerodynamics using a monolithic overset approach," *Computational Mechanics*, 77, 1245–1265, 2026. https://doi.org/10.1007/s00466-025-02705-2.

[10] T. G. Myers and S. L. Mitchell, "A mathematical analysis of the motion of an in-flight soccer ball," *Sports Engineering*, 16, 29–41, 2013. https://doi.org/10.1007/s12283-012-0105-8.

[11] T. Asai, K. Seo, O. Kobayashi and R. Sakashita, "Fundamental aerodynamics of the soccer ball," *Sports Engineering*, 10, 101–110, 2007. https://doi.org/10.1007/BF02844207.

[12] A. L. Kiratidis and D. B. Leinweber, "An aerodynamic analysis of recent FIFA World Cup balls," *European Journal of Physics*, 39, 034001, 2018. https://doi.org/10.1088/1361-6404/aaa888. Equations and parameters: https://arxiv.org/abs/1710.02784v2.

[13] Ł. Grad, K. M. Czajkowski and A. Varashylau, "Physics-Based 3D Ball Trajectory Reconstruction from Monocular Soccer Video: A Multi-Model Benchmark," *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*, 9940–9948, 2026. Supplementary material, Section 8.4.

[14] MuJoCo developers, MuJoCo 3.8.1, ellipsoid fluid model, `mj_viscousForces` in `src/engine/engine_passive.c`. Version 3.8.1 source; default angular drag: XML reference, geom/fluidcoef. Accessed 8 November 2026.

[15] J. E. Goff and M. J. Carré, "Trajectory analysis of a soccer ball," *American Journal of Physics*, 77, 1020–1027, 2009. https://doi.org/10.1119/1.3197187.

[16] F. Alam, H. Chowdhury, M. Stemmer, Z. Wang and J. Yang, "Effects of surface structure on soccer ball aerodynamics," *Procedia Engineering*, 34, 146–151, 2012. https://doi.org/10.1016/j.proeng.2012.04.026.

Appendix A: Data Sources, Frame Selection and Parameter Bounds

The three penalties come from the Sportec Open DFL dataset, which associates match events with TRACAB Gen5 multi-camera tracking at 25 Hz[6, 7]. The supplied CSV retains 39 source frames, with 15, 12 and 12 frames for the three events.

Table 9: Correspondence between case labels and source records

Case	Event ID	Player	Selected samples
Placement 1	J03WMX	Dejan Ljubicic	3, 4, 5
Placement 2	J03WR9	Philipp Klement	3, 4, 5
Power shot	J03WOH	Dawid Igor Kownacki	2, 3, 4

The first selected position is the time origin and the fitting times are 0, 0.04 and 0.08 s. Their source-time origins are 0.08, 0.08 and 0.04 s in the table's order. Repeated and post-goal records are retained; later positions are used only for validation. The observed goal crossing uses a source frame or interpolation. The height field is interpreted in metres as ball-centre height, and coordinates are converted with $x = X - 41.5$, $y = Y$ and $z = Z_{\text{original}}$.

Table 10 gives the early inputs and goal crossings; Table 11 gives the inversion ranges.

Table 10: Selected early positions and observed goal crossings in local coordinates.

Event	Source frame	t (s)	x (m)	y (m)	z (m)
Placement 1	152602	0.00	3.040	0.550	0.130
Placement 1	152603	0.04	4.190	0.820	0.150
Placement 1	152604	0.08	5.250	1.070	0.180
Placement 1 goal	152610	0.3200	11.0000	2.7500	0.1600
Placement 2	176245	0.00	2.570	0.850	0.340
Placement 2	176246	0.04	3.640	1.200	0.410
Placement 2	176247	0.08	4.710	1.530	0.470
Placement 2 goal	176253–176254	0.3295	11.0000	3.0697	0.4657
Power shot	128955	0.00	0.770	0.080	0.240
Power shot	128956	0.04	1.990	0.340	0.460
Power shot	128957	0.08	3.180	0.620	0.650
Power shot goal	128964–128965	0.3643	11.0000	2.9188	1.1789

Table 11: Component bounds and force-correction settings for inversion.

Quantity	Search interval
v_{x0} (M1, M2, M4)	$[0, 35]$ m/s
v_{y0}, v_{z0} (M1, M2, M4)	$[-35, 35]$ m/s
$\|\boldsymbol{\omega}_0\|$ (M4)	30 rad/s
Azimuth, elevation (M4)	$[-\pi, \pi], [-\pi/2, \pi/2]$ rad
$C_D/C_D^*, \alpha/\alpha^*$	$[0.5, 1.5]$

Table 12 gives the fitted M4 parameters, with $\boldsymbol{n} = \boldsymbol{\omega}/30$.

Table 12: Fitted M4 states at the selected starting frames.

Event	v_0 (m/s)	C_D	α	n_x	n_y	n_z
J03WMX	29.20	0.3623	2.2877	0.2280	-0.9728	0.0413
J03WR9	28.34	0.1208	2.2877	0.1439	-0.2711	-0.9517
J03WOH	32.05	0.3238	2.2877	-0.2276	0.3170	0.9207

Reproduction Code

The project contains the original penalty CSV, coordinate conversion, shared inversion, prediction results and figure scripts. The excerpts show the common state equation and force evaluation. Paper and program use the same coefficients and bounds; computation, plotting and bilingual table generation are separate commands.

```
--
def state_derivative(t, state, config):
    """The single nine-state RHS used for prediction and inverse fitting."""
    f = forces(state[3:6], state[6:9], config)
    return np.r_[state[3:6], sum(f.values())/config.mass,
                rotational_torque(state[6:9], config)/config.moment_of_inertia]
```

```
--
def forces(velocity, spin, config):
    """Return force components on ball, in newtons."""
    v = np.asarray(velocity, dtype=float) - np.asarray(config.wind, dtype=float)
    omega = np.asarray(spin, dtype=float)
    speed = np.linalg.norm(v)
    weight = (config.mass * np.asarray(config.constant_acceleration, dtype=float)
              if config.constant_acceleration is not None else
              np.array([0., 0., -config.mass * config.gravity if config.use_gravity else
                  0.]))
    drag = np.zeros(3)
    magnus = np.zeros(3)
    if speed > 1e-12:
        cd, cl, _, _, _ = coefficients(velocity, spin, config)
        q_area = 0.5 * config.air_density * np.pi * config.radius**2 * speed**2
        if config.use_drag:
            drag = -q_area * config.drag_scale * cd * v / speed
        cross = np.cross(omega, v)
        norm = np.linalg.norm(cross)
        if config.use_magnus and norm > 1e-12:
            # The literature fit is for transverse spin. Project its effect
            # onto the plane perpendicular to airflow for an oblique spin axis.
            sin_angle = norm / (np.linalg.norm(omega) * speed)
            magnus = q_area * config.lift_scale * cl * sin_angle * cross / norm
    return {"gravity": weight, "drag": drag, "magnus": magnus}
```